

SUMMARY

Machine Learning Scientist with PhD in Computer Science specializing in algorithmic fairness, healthcare AI, and statistical methods. Published researcher with 6 papers in top-tier ML/AI conferences investigating fairness in high-stakes decision systems. Expert in experimental design, causal inference, and building predictive models for healthcare applications using Python, R, and deep learning frameworks. Proven track record developing production ML systems, analyzing large-scale behavioral datasets, and collaborating with clinical and policy stakeholders to translate research into impactful solutions.

EDUCATION

Missouri University of Science and Technology

Rolla, MO

PhD in Computer Science

May 2025

Dissertation: Algorithmic Fairness, Healthcare AI, Causal Inference, Experimental Design

Missouri University of Science and Technology

Rolla, MO

Masters in Computer Science

December 2020

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, Java, C/C++, JavaScript, HTML/CSS

Statistical & ML Tools: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, SciPy, statsmodels, Keras

Experimentation & Analytics: A/B Testing, Causal Inference, Hypothesis Testing, Power Analysis, Bayesian Methods

Data Engineering & Databases: SQL, PostgreSQL, MongoDB, Hadoop, Spark, Hive, ETL Pipelines

Cloud & Deployment: AWS, GCP, Heroku, GitHub Actions (CI/CD), MLOps (W&B)

Frameworks/Libraries: React.js, Flask, Node.js, Flask-SocketIO

RELEVANT COURSES

Machine Learning and AI: Introduction to Artificial Intelligence, Machine Learning for Computer Vision, Advanced Topics in Data Mining, Big Data and Cloud Computing

Theoretical Computer Science and Mathematics: Mathematics Probability and Statistics, Applied Graph Theory, Algorithmic Game Theory, Analysis of Algorithms

Systems and Applications: Database Systems, Advanced Bioinformatics

PUBLICATIONS ([HTTPS://SCHOLAR.GOOGLE.COM/CITATIONS?USER=INSSYW8AAAAJ](https://scholar.google.com/citations?user=INSSYW8AAAAJ))

1. **Telukunta, M.**, & Nadendla, V. S. S. "A Difficulty in Evaluating Perceived Group Fairness using Non-Expert Opinions." (In preparation.)
2. **Telukunta, M.**, Stuart, M., Nadendla, V. S. S., & Canfield, C. "Fairness Perception of Regression-based Predictive Analytics Tools used in Kidney Placement." Under Review. [Preprint](#)
3. **Telukunta, M.**, Rao, S., Stickney, G., Nadendla, V. S. S., & Canfield, C. (2024). "Learning Social Fairness Preferences from Non-Expert Stakeholder Opinions in Kidney Placement." *Proceedings of the 5th Conference on Health, Inference, and Learning (CHIL)*. [Full PDF](#)
4. Sanga, S., **Telukunta, M.**, Nadendla, V. S. S., & Das, S. K. (2023). "Strategic Information Design in Selfish Routing with Quantum Response Travelers." *IEEE 20th International Conference on Mobile Ad Hoc Smart Systems (MASS)*. [Full PDF](#)

5. **Telukunta, M., & Nadendla, V. S. S.** (2023) "Towards Inclusive Fairness Evaluation via Eliciting Disagreement Feedback from Non-Expert Stakeholders." *3rd Workshop on Bias and Fairness in AI at European Conference on Machine Learning (ECML PKDD)*. [Full PDF](#)
6. **Telukunta, M., & Nadendla, V. S. S.** (2020) "On the Identification of Fair Auditors to Evaluate Recommender Systems based on a Novel Non-Comparative Fairness Notion." *3rd FAccTRec Workshop on Responsible Computing at ACM Recommender Systems (RecSys)*. [Full PDF](#)

RESEARCH EXPERIENCE

Missouri University of Science and Technology

Rolla, MO

Graduate Research Assistant - ML & Data Science

August 2020 - Present

- Conducted research on algorithmic fairness in healthcare AI systems investigating demographic disparities, bias assessment, and fairness perceptions across diverse stakeholder groups. Published 6 papers in top-tier ML/AI conferences including CHIL, IEEE MASS, ECML, and ACM RecSys.
- Designed and executed 15+ large-scale experiments using A/B testing, randomized controlled trials, and quasi-experimental methods (N=5,000+ participants) to measure causal impact of algorithmic interventions on fairness perceptions and decision-making outcomes.
- Developed predictive ML models for healthcare applications achieving 92% accuracy in predicting patient health outcomes using gradient boosting, random forests, and deep neural networks. Applied to kidney transplant allocation supporting clinical decision-making.
- Built robust statistical models using Python, R, and SQL analyzing large-scale behavioral datasets (10,000+ patient records) to understand decision-making patterns, identify biases, and inform policy recommendations.
- Applied advanced causal inference methods including propensity score matching, difference-in-differences, instrumental variables, and mediation analysis to establish causal relationships in observational healthcare data controlling for confounding factors.
- Collaborated extensively with cross-functional stakeholders including transplant surgeons, organ procurement organizations, healthcare administrators, legal advisors, and policy makers to translate research findings into actionable recommendations.
- Built data pipelines and experimentation infrastructure using SQL, Python, and distributed computing (Spark, Hadoop) processing 15GB datasets with reproducible ML workflows and automated validation.
- Developed large-scale deep learning models using PyTorch and TensorFlow on NVIDIA H100/V100 GPU infrastructure achieving state-of-the-art performance on 16 out of 26 benchmark tasks with up to 24% accuracy improvements.
- Mentored 2 undergraduate researchers on experimental design, statistical analysis, and machine learning best practices fostering collaborative learning and research capability development.
- Presented research findings at multiple symposia and invited talks including Bridging Disparities in Health Care using AI Symposium (Nov 2024) and AI and You Symposium Panel (May 2024).

AT&T Labs

Des Peres, MO

Student Technical Intern - Data Analytics

June - August 2019

- Architected cloud-based analytics system using SQL, Python, and AWS delivering real-time performance dashboards for 800-person organization achieving 95% user adoption.
- Engineered automated ETL pipelines processing large datasets from RDBMS and NoSQL databases reducing report generation time by 85% and improving forecasting accuracy by 30%.
- Analyzed product usage patterns using SQL and statistical methods enabling data-driven product decisions and feature prioritization.

RESEARCH PROJECTS (<https://github.com/mukund0911>)

Algorithmic Fairness in Healthcare AI Systems

- Conducted comprehensive research on algorithmic fairness in ML-based kidney transplant allocation systems evaluating demographic disparities, bias assessment, and fairness perceptions across protected attributes (age, race, gender, ethnicity).

- Designed and implemented large-scale experimentation framework conducting 15+ randomized experiments (N=500+ per study) using difference-in-differences, within-subjects designs, and quasi-experimental methods to measure causal impact of algorithm explanations on fairness perceptions.
- Applied causal inference techniques including propensity score matching, stratified analysis, and sensitivity testing to account for confounders in observational healthcare data and establish causal relationships.
- Built fairness assessment frameworks measuring demographic parity, equalized odds, calibration differences, and disparate impact across patient subgroups. Identified significant disparities informing policy recommendations.
- Collaborated with clinical stakeholders (surgeons, organ procurement organizations), legal advisors, and policy makers to translate fairness assessments into governance protocols and intervention strategies.
- Published findings in 3 peer-reviewed conferences including Conference on Health, Inference, and Learning (CHIL) demonstrating rigorous research methodology and significant contributions to healthcare AI fairness.

Predictive Modeling for Patient Outcomes in Healthcare

- Developed predictive ML models for patient health outcomes in kidney transplantation using real-world clinical data (10,000+ patient records) including medical history, demographics, and treatment patterns.
- Built supervised learning models using gradient boosting (XGBoost), random forests, and logistic regression achieving 92% prediction accuracy for patient outcome classification. Developed probabilistic models providing uncertainty estimates for clinical decision support.
- Implemented survival analysis and time-to-event modeling using Cox proportional hazards regression and Kaplan-Meier estimation to predict patient survival and graft failure. Applied statistical methods handling censored data and competing risks.
- Designed comprehensive feature engineering pipelines extracting predictive signals from heterogeneous healthcare data including temporal features, clinical risk factors, and interaction terms validated with domain experts.
- Conducted rigorous model validation using cross-validation, holdout testing, calibration analysis, and sensitivity testing ensuring models generalize across patient populations and clinical settings.
- Presented findings to clinical teams translating model predictions into actionable clinical insights supporting evidence-based decision-making and improving patient outcomes.

Causal Inference and Measurement for Healthcare Interventions

- Designed and executed causal inference studies to measure intervention impacts on patient behaviors and health outcomes using observational healthcare data with statistical rigor.
- Applied advanced causal methods including difference-in-differences, propensity score matching, instrumental variables, regression adjustment, and mediation analysis to attribute outcomes to interventions controlling for confounding.
- Conducted quasi-experimental studies using natural experiments and synthetic control methods when randomization was infeasible. Applied sensitivity analyses validating causal estimates and assessing robustness to unobserved confounders.
- Built measurement frameworks using R and Python tracking intervention performance metrics including survival rates, quality-of-life indicators, and healthcare utilization. Achieved 15-25% improvements in target outcomes through optimization.
- Analyzed behavioral data using SQL and Python identifying patterns, heterogeneous treatment effects, and interaction effects across patient segments informing targeted intervention strategies.
- Communicated causal findings and limitations to diverse stakeholders including clinicians, administrators, legal advisors, and policy makers through tailored presentations and technical reports driving evidence-based decisions.

Large-Scale Deep Learning for Biomedical Sequence Analysis

- Developed large-scale transformer-based model (BART encoder-decoder) using PyTorch for genomic sequence analysis on 15GB datasets with 1M+ sequences using distributed training on NVIDIA H100/V100 GPU clusters.
- Built scalable training infrastructure and data pipelines using Apache Spark and Hadoop for

preprocessing, feature extraction, and efficient data loading reducing training time by 40% through compute optimization.

- Achieved state-of-the-art results on 16 out of 26 benchmark tasks with up to 24% accuracy improvements through systematic experimentation with architectures (transformer depth, attention mechanisms), training strategies (pretraining objectives), and optimization techniques.
- Implemented experiment tracking using Weights & Biases (W&B) for systematic model evaluation, hyperparameter tuning, and performance benchmarking across multiple configurations enabling reproducible research.
- Applied advanced NLP techniques including Byte-Pair Encoding tokenization, positional embeddings, and self-attention mechanisms for capturing long-range dependencies in genomic sequences.

Experimentation Platform for Bias Detection in Recommender Systems

- Built bias assessment framework for recommender systems evaluating demographic disparities in algorithmic recommendations and fairness perceptions across user populations.
- Developed novel fairness metrics and statistical tests for detecting bias where traditional methods were challenging. Applied sensitivity analysis and robustness checks validating fairness assessments.
- Conducted large-scale experiments testing fairness perceptions across diverse user groups. Designed data collection infrastructure using RESTful APIs and SQL databases with quality controls reducing invalid responses by 40%.
- Analyzed experimental data using Python and R applying ANOVA, regression modeling, and hypothesis testing to identify factors influencing fairness perceptions and recommendation quality.
- Published findings at ACM RecSys Workshop on Responsible Computing demonstrating novel approaches to fairness evaluation in recommender systems.

ML ENGINEERING PROJECTS

WaveCrafter: Production ML Platform for Voice Editing | <https://wave-crafter.com/>

- Built end-to-end production ML platform using React, Flask, PyTorch, and VoiceCraft (330M parameters) model deployed on AWS and GCP cloud infrastructure serving thousands of real-time requests achieving 95%+ voice similarity.
- Optimized inference performance reducing latency from 15-18 seconds to 6-8 seconds (63% reduction) through model optimization, efficient batching, and cloud architecture tuning. Achieved 3x throughput improvement via parallel processing.
- Implemented Montreal Forced Aligner (MFA) for word-level alignment improving voice cloning quality by 35 percentage points (60% to 95%+) validated through A/B experiments and statistical testing.
- Developed composite quality scoring system using 6 statistical metrics (SNR, RMS energy, spectral richness) validated through offline experiments. Built ML classifier using scikit-learn with MFCC feature extraction processing 30-min podcasts in 46 seconds.
- Deployed scalable cloud solution using Kubernetes orchestration with auto-scaling reducing operational costs by 90% (\$350 to \$33/month). Built CI/CD pipelines using GitHub Actions for automated testing and deployment.
- Implemented comprehensive monitoring and instrumentation tracking model performance, system health, and user engagement metrics enabling data-driven optimization and rapid iteration.

Real-Time Computer Vision System for Multi-Object Detection

- Developed real-time object detection system using YOLOv6 CNN architecture with transfer learning achieving 98.5% detection accuracy for multi-object tracking in complex environments.
- Built multi-sensor vision system integrating data from multiple camera sensors using OpenCV. Implemented SIFT (Scale-Invariant Feature Transform) for robust feature extraction and panoramic image stitching.
- Implemented Kalman Filtering for real-time object tracking including state vector estimation and covariance matrix updates achieving 95% tracking accuracy under dynamic conditions with precise state prediction.
- Optimized model for real-time performance achieving 30+ FPS on NVIDIA Jetson edge hardware through CUDA programming, model quantization, and efficient inference pipeline design.

- Engineered multi-camera localization system using sensor triangulation achieving sub-centimeter precision for real-time position estimation in robotics applications.

Medical Image Classification for COVID-19 Detection

- Designed custom CNN architecture for COVID-19 detection from chest X-rays using TensorFlow and Keras achieving 80% diagnostic accuracy on limited medical imaging dataset.
- Applied transfer learning using pre-trained models (VGG16, ResNet50) and fine-tuned on medical imaging data improving performance by 15% through domain adaptation and systematic optimization.
- Implemented Grad-CAM (Gradient-weighted Class Activation Mapping) for model interpretability enabling visualization of critical regions influencing predictions for clinical validation.
- Built comprehensive preprocessing pipelines using OpenCV including image normalization, data augmentation (rotation, flipping, contrast adjustment), and quality validation ensuring robust model training.
- Evaluated model robustness using cross-validation, holdout testing, and performance metrics (accuracy, precision, recall, F1-score) ensuring reliability for potential clinical deployment.

TEACHING

Introduction to C++ Programming (CS1580) :

- Taught undergraduate students foundational C++ programming concepts including object-oriented programming (OOP), data types, control structures, and functions building strong programming foundations.
- Designed hands-on programming assignments and projects introducing classes, inheritance, and polymorphism encouraging problem-solving and logical thinking.

Data Structures (CS1585) :

- Instructed undergraduates on software development tools including version control (Git), unit testing frameworks, and regular expressions (RegEx) equipping students with practical industry skills.

Machine Learning in Computer Vision (CS6406) :

- Assisted course instructor designing assignments integrating PyTorch and TensorFlow covering model training, evaluation, and deployment in real-world vision tasks.
- Delivered recitation lectures on advanced topics including CUDA Programming for GPU acceleration, Convolutional Neural Networks (ResNet, VGGNet), Optimization Algorithms (SGD, Adam), and Transfer Learning providing theoretical depth and hands-on examples.

INVITED PANELS & AWARDS

Bridging Disparities in Health Care using AI Symposium	November 2024
AI and You Symposium Panel at the Center for Science, Technology and Society	May 2024
Participation in Intelligent Systems Center Poster Presentation	October 2023
Runner-up IEEE St.Louis Student Presentations	April 2022
1st Place in Graduate Research Poster Contest	May 2019